Big Data Analytics Project

# Big-Data Driven Taxi Fare Prediction & Demand Hotspot Analytics


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# Foundations: Problem, Objectives & Big Data Need

## 1. Problem Statement

- **Urban mobility:** 175M+ taxi hailing events with high volatility across 263 NYC zones.
- **Price uncertainty:** Fares change with congestion, tolls, trip kinematics, and weather.
- **Imbalance:** Drivers cluster in hubs while outer boroughs remain underserved.

## 2. Project Objectives

- **Fare prediction:** Ridge Regression trained on 26.8M historical trips.
- **Demand regimes:** K-Means ($K = 3$) for Low, Moderate, and High-demand hotspots.
- **Streaming:** Structured Streaming with 3-sec micro-batches & 1-hr sliding window.

## 3. Why Big Data?

- 175M+ raw trips exceed single-node memory and processing limits.
- 4.63M spatiotemporal bins require distributed shuffle and aggregation.
- Real-time scoring needs stateful, fault-tolerant execution engines.

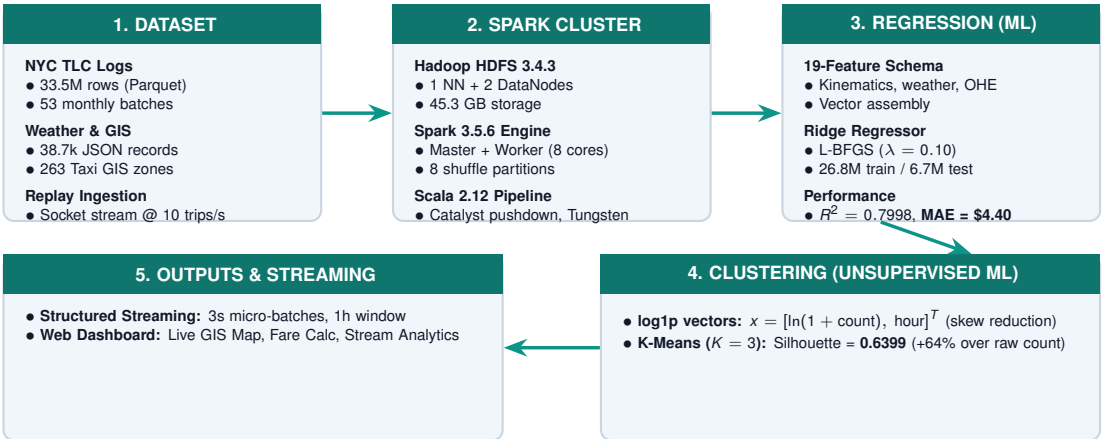
# Dataset: Source, Size and the Five V's

| Layer | Source | Format | Rows | Size | Period |
|---|---|---|---|---|---|
| **Structured** | NYC TLC Yellow & Green logs | Snappy Parquet | 33,484,865 | 2.9 GB raw / 4.9 GB | 2022-01 – 2026-05 |
| **Semi-struct.** | Open-Meteo weather reanalysis | Nested JSON | 38,712 | 18.4 MB | 2022-01 – 2026-05 |
| **Spatial** | NYC Taxi Zones GIS shapefiles | CSV / GeoJSON | 263 zones | 1.2 MB | Static |

| V | Highlight | Description |
|---|---|---|
| **Volume** | **33.5M rows** | 128 MB HDFS blocks with 2× replication for parallel access. |
| **Variety** | **Multi-Format** | Columnar trips, JSON weather, and GIS spatial zones unified in Spark schemas. |
| **Velocity** | **3s Stream** | Structured Streaming with 1-hour sliding windows continuously updating demand. |
| **Veracity** | **Cleaned Data** | Filtered 0.1–100 mi and \$3–\$500 fares; chronological split prevents leakage. |
| **Value** | $R^2 = 0.7998$ | \$4.40 MAE transparent fare pricing; hotspot analysis guides fleet routing. |

**Key attributes:** Pickup/Dropoff datetimes, Distance, Duration, Speed, Passenger Count, LocationIDs, Boroughs, Temp, Precip, Wind.

# System Architecture: End-to-End Pipeline



# Big Data Execution: Spark Engine Optimizations

## DataFrames, Catalyst & DAG Optimizations

- Built using DataFrame-based `spark.ml` API for optimized Catalyst logical plan generation.
- **Predicate Pushdown:** Filters unneeded partition files directly at the Parquet storage layer.
- **Projection Pruning:** Loads only active 19 model features into JVM off-heap memory.
- **Tungsten Engine:** Off-heap binary memory management eliminates GC and object creation overhead.

## Partitioning & Memory Management

- Set `spark.sql.shuffle.partitions = 8` to align with cluster CPU cores.
- Strategic caching with `demandDF.cache()` across iterative $K$-means parameter grids.
- Cached prediction DataFrames prevent re-evaluating upstream ML pipelines during error slicing.
- Explicit `unpersist()` calls release worker executor memory after iteration.

# Scala Engineering: Production Code Snippets (1/2)

## Project Structure

```
surgesense-project/
|-- build.sbt
`-- src/main/scala/surgesense/
    |-- Phase1SplitAndEDA.scala
    |-- FinalModelFeatures.scala
    |-- FinalLinearRegression.scala
    |-- DemandHotspotsV2.scala
    |-- FareErrorAnalysis.scala
    `-- StreamingInference.scala
```

**Compile:** `sbt package`

**Submit:**

```
spark-submit --master
  spark://spark-master:7077
  --deploy-mode client
```

**Execution:** Standalone cluster mode with 8 shuffle partitions.

## 1. Regularized Linear Regression Pipeline

```
// FinalLinearRegression.scala
val lr = new LinearRegression()
  .setLabelCol("fare_amount")
  .setFeaturesCol("scaled_features")
  .setMaxIter(50)
  .setRegParam(0.10)    // L2 (Ridge)
  .setSolver("l-bfgs") // quasi-Newton

val lrModel = lr.fit(trainDF)

lrModel.write.overwrite()
  .save("hdfs://namenode:9000/
         surgesense/models/...")
```

# Scala Engineering: Production Code Snippets (2/2)

## 2. Stateful Sliding Window Stream

```
// StreamingInference.scala
val windowedStream = rawStream
  .withWatermark("pickup_datetime",
                 "10 minutes")
  .groupBy(
    window(col("pickup_datetime"),
           "1 hour", "5 minutes"),
    col("pickup_zone"),
    col("pickup_borough")
  )
  .agg(count("*").cast("double")
    .alias("pickup_count"))
```

## 3. K-Means Inference

```
val clustered = kmeansModel
  .transform(scaledStream)
  .withColumn(
    "demand_tier",
    when(col("cluster") === 2,
         "HIGH-DEMAND HOTSPOT")
      .when(col("cluster") === 0,
            "MODERATE DEMAND")
      .otherwise("LOW DEMAND")
  )
```

# Spark ML Pipeline: Model Architecture & Tuning

## Algorithm Specs (LR)

| Parameter | Value |
|---|---|
| Solver | `l-bfgs` |
| RegParam ($\lambda$) | **0.10** |
| ElasticNet ($\alpha$) | 0.0 |
| MaxIter | 50 |
| Standardization | True |

$\alpha = 0.0$ corresponds to Ridge ($L_2$) regularization.

## Hyperparameter Tuning ($\lambda$)

| Metric | $\lambda = 0.01$ | $\lambda = 0.10$ |
|---|---|---|
| $R^2$ Score | 0.79958 | **0.79982** |
| RMSE | \$7.7935 | **\$7.7889** |
| MAE | \$4.4118 | **\$4.4027** |
| $\pm$\$2 Tol. | 26.07% | **26.30%** |

Evaluated on 6.7M hold-out test rows (Mar–May 2026). Tuned $\lambda = 0.10$ achieves optimal bias-variance tradeoff.

## Hotspot Clustering (K-Means)

| Clustering Spec | Config / Metric |
|---|---|
| Vector Space | $\mathbf{x} = [\ln(1 + \text{pcnt}), \text{hr}]^T$ |
| Search Range | $K \in [2, 8]$ |
| Optimal $K$ | $K = 3$ |
| Silhouette Score | **0.6399** (+64%) |

Identifies 3 demand tiers: Low, Moderate and High-Demand Hotspots.

# Results: Model Evaluation & Comparison (Baseline vs. Tuned)

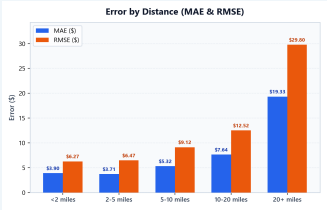
## Linear Regression: Baseline vs. Tuned Metrics

| Metric | Original ($\lambda$=0.01) | Tuned ($\lambda$=0.10) |
|---|---|---|
| $R^2$ Score | 0.799582 | **0.799821** |
| RMSE | \$7.7935 | **\$7.7889** |
| MAE | \$4.4118 | **\$4.4027** |
| Tolerance ($\pm$\$2) | 26.0683% | **26.3034%** |

## Key Performance & Business Insights

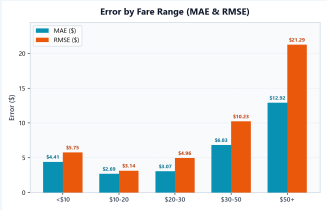
- L2 regularization with $\lambda = 0.10$ achieves the highest variance explained ($R^2 \approx 80\%$) across 6.7M test trips.
- Mean Absolute Error is constrained to \$4.40 citywide, giving drivers and riders reliable estimates.
- Over 1.76M predictions (26.30%) fall within $\pm$\$2.00 of the metered fare.
- Lower RMSE confirms effective mitigation of large-penalty outliers through L-BFGS convergence.

## Error by Distance Slice (MAE vs. RMSE)



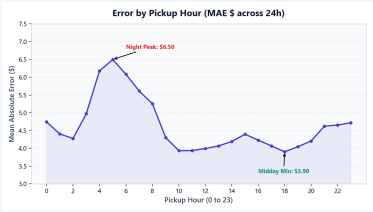

- Minimum error at MAE $3.71 across core volume (2–5 miles, 85% of rides).
- Error rises to MAE $19.33 on long-haul trips (>20 miles) due to dynamic airport tolls.

## Error by Fare Tier (MAE vs. RMSE)



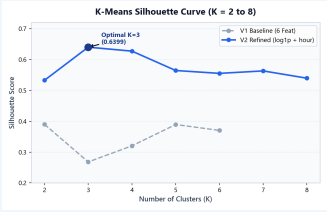

- Strongest fit at MAE $2.69 for standard $10–$20 fares.
- Higher RMSE of $21.29 on $50+ rides reflects long-distance surcharge variance.

## Error by Pickup Hour (MAE $ across 24h)



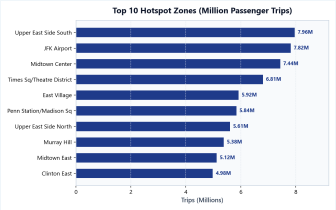

- Stable midday traffic (10:00–14:00) achieves the lowest error at $3.94.
- Early morning at 05:00 shows higher dispersion of $6.50 during fleet shift handover.

## K-Means Silhouette Curve (V1 vs. V2 Refined)



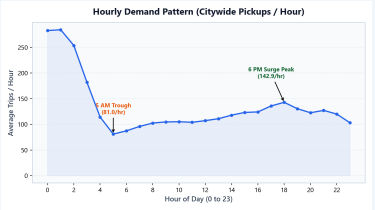

- The improved k - means boosts silhouette score from 0.3898 to 0.6399.
- A distinct global optimum at $K = 3$ supports 3 demand regimes.

## Top 10 Hotspot Zones (Million Trips)



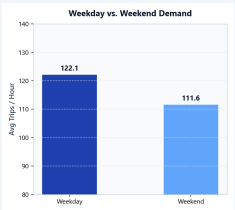

- Highest concentrations: Upper East Side South (7.96M) and JFK Airport (7.82M).
- Major commercial transit points include Midtown Center (7.44M) and Times Square (6.81M).

## 24-Hour Citywide Demand Pattern



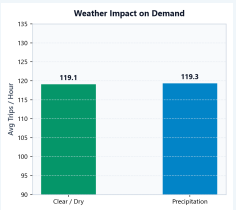

- Citywide demand rises from a 05:00 trough (81.0/hr) steadily through the workday.
- Evening surge peaks at 142.9/hr at 18:00 during commercial rush-hour exit.

## Weekday vs. Weekend Demand (Trips / Hour)




- Weekday volume of 122.1/hr is driven by repetitive Manhattan business transit.
- Weekend volume of 111.6/hr shifts toward leisure corridors and airports.

## Weather Impact on Demand (Trips / Hour)




- Rain causes pedestrian modal substitution, maintaining trip rate at 119.3/hr.
- Overall demand remains inelastic at 119.1 vs 119.3/hr across weather conditions.

# Limitations & Future work

## Technical Challenges & Limitations

- **Long-haul variance:** 20+ mile trips see elevated error ( **MAE $19.33** ) due to unobserved toll pricing.
- **Target leakage:** Post-trip variables such as `tolls_amount` cannot be used for pre-ride estimation.
- **Event-time skew:** Requires watermarking and state eviction to prevent unbounded state growth.

## Conclusion & Phase 2 Roadmap

- Processed 33.5M records on Apache Spark and Hadoop HDFS.
- Linear fare regression achieved $R^2 = 0.7998$ and **MAE $\approx$ $4.40** .
- $K$-Means clustering achieved an optimal silhouette score of **0.6399** .
- **Phase 2:** Integrate toll engine, hierarchical fare models, and Apache Kafka streaming ingest.

# Citations & Project Repository

## References & Data Sources


**[1] NYC Taxi & Limousine Commission (TLC)**, *TLC Trip Record Data*.
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**[3] K. Sundravadivelu et al.**, "Taxi Fare Prediction using Random Forest: A Data-Driven Approach," *Proc. ICISS*, 2025.
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**[4] S. Agarwal et al.**, "Taxi Fare Prediction Using Various Machine Learning Models," *IJFMR*, 2025.
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**[5] T. Ramadhan et al.**, "Application of K-Means Clustering for Urban Transportation Pattern Analysis," *JAIEA*, 2026.
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**[6] S. Mishra and S. Khetarpaul**, "Identifying and Recommending Taxi Hotspots in Spatio-Temporal Space," *GeoInformatica*, 2025.
https://link.springer.com/article/10.1007/s10707-024-00524-w


## Project Repository & Submission Link

**GitHub Repository:** https://github.com/Vivethasri985/bigdata-taxi-fare-prediction